E1 dry run: the known-log base against the abscissa base

SCRATCH, not data. This is the PLAN §5 step 2 dry run. `PROTOCOL.md` is not frozen, the host is L0 (no CPU pinning, no isolation lock, other load allowed), and nothing here is a result of record. Operation counts and group-addition equivalents (GAE) are exact and deterministic; wall time is not used anywhere. **No security claim.** The known-log base is a generic precomputation table in the Bernstein–Lange shape. No exponent moves.

What ran

item	value
curves	two 24-bit prime-order curves, `a = −3`, cofactor 1, found by `examples/a_minus_3_prime_order_curves.rs` (seed 20261019); generator = smallest-abscissa point
c1	`p = 8582641`, `b = 6723661`, `r = 8580037` (2^23.03), B = 205
c2	`p = 13903607`, `b = 5849518`, `r = 13902853` (2^23.73), B = 240
B	`ceil(r^(1/3))`, the family optimum of PLAN §4
arms	`ic-abscissa` (`prime-abscissa:size=B`), `ic-knownlog` (`known-log:size=B`), `rho-neg`; both IC arms `mitm:negation_folded=1,m=2`, walk targets
targets	16 `planted_shared` per curve, 2 measured rounds after 1 warm-up, so 32 runs per arm per curve, all verified (288 of 288 records including warm-up)
sessions	c1 `ECBS1hdfb4b2694578`, c2 `ECBS1h3e4cb8782f3c`
specs, records	`dryrun/spec-c1.json`, `dryrun/spec-c2.json`, `dryrun/records/`
aggregation	`examples/generator_effect_dryrun_summary.rs` writes `dryrun/summary.json`; 10 000 bootstrap resamples, percentile 90% intervals, seed 20261019
graph	`dryrun/dryrun.svg`, source `dryrun/plot_dryrun.py` (plots recorded values only)

Pricing (AGENTS.md §5a). The pinned calibration prices group additions only. Lookups, target-guard probes, square roots, Legendre symbols and `row_ops` have no pinned ratio on these fresh curves, so they are priced at $w \in \{0, 0.1, 1\}$ GAE per operation and every number is shown at all three. S = GAE / √r. At `w = 0` the abscissa base's construction and its linear algebra cost exactly 0 GAE, because they are only unpriced native work.

Result 1: the baseline's set-up and linear-algebra share is small

Share of the `ic-abscissa` total S (measured counts, derived shares).

curve	w	total S	base construction	linear algebra	**set-up + LA**	pair-table build	target draw + probes
c1	0	19.14	0	0	**0.0%**	75.3%	24.6%
c1	0.1	20.03	0.11%	0.24%	**0.35%**	72.0%	27.6%
c1	1	28.07	0.76%	1.73%	**2.5%**	51.4%	46.1%
c2	0	20.17	0	0	**0.0%**	76.9%	23.1%
c2	0.1	21.07	0.09%	0.19%	**0.29%**	73.6%	26.0%
c2	1	29.10	0.68%	1.40%	**2.1%**	53.3%	44.6%

The cost of the baseline at this size is the pair table (42 230 additions on c1) and the walk that feeds it, not the base's algebra. Removing set-up and LA entirely can save at most **0% to 2.5%** of S. Rho on the same curves: S = 1.13 / 1.24 / 2.14 on c1 and 0.95 / 1.04 / 1.81 on c2 at w = 0 / 0.1 / 1, so both IC arms sit at about 13 to 21 times rho at T = 1 (derived).

Result 2: the online yield is about 100 times lower, not equal

Relations found per target tried, pooled over 32 runs per arm, 90% bootstrap interval.

curve	abscissa	known-log	**ratio known-log / abscissa**
c1	9.75e-3 [9.45e-3, 1.006e-2]	9.43e-5 [7.32e-5, 1.266e-4]	**0.0097 [0.0074, 0.0130]**
c2	8.25e-3 [8.04e-3, 8.46e-3]	7.61e-5 [5.93e-5, 1.032e-4]	**0.0092 [0.0072, 0.0126]**

PLAN §4 E1 predicted a ratio of 1.00 with a 90% interval inside [0.97, 1.03]. **That prediction is false, and structurally so.** The base `{±[i]G}` is an arithmetic progression, so every sum of two of its members is one of `±i ± j`, an integer multiple of G in `[−2B, 2B]`. The pair table holds at most `4B + 1` distinct sums (821 on c1, 961 on c2), where an unstructured base of B points holds about `C(2B, 2)` (about 84 000 on c1). The oracle tests membership of a *sum*, and sums of a progression collide. The predicted hit probability per target, `(4B + 1) / r`, is 9.57e-5 on c1 and 6.91e-5 on c2 (derived), against 9.4e-5 and 7.6e-5 measured, both inside the interval. PLAN §1D and §2 said that the MITM oracle makes the base "an arbitrary set of B points". It does not: it makes the base's *sumset* the object, and the sumset of `F_G(B)` has size O(B), not O(B^m).

Result 3: total S at T = 1 is a tie, by coincidence of trial counts

Total S of `ic-knownlog` over `ic-abscissa`, paired by workload and round.

curve	w = 0	w = 0.1	w = 1
c1	0.973 [0.922, 1.030]	0.965 [0.908, 1.030]	0.921 [0.819, 1.031]
c2	0.968 [0.917, 1.022]	0.960 [0.902, 1.022]	0.913 [0.809, 1.021]

Every interval contains 1. Under PLAN §4′s decision rule (interval below 1 at every weight) this is **"accounting, no improvement"** at T = 1. The ratio is not a saving from removed algebra (Result 1 caps that at 2.5%). It is two trial counts that happen to be close: the abscissa run collects about 118 relations at 1% each, about 12 000 targets; the known-log run needs one relation at 0.01% each, about 10 600 targets. The pair-table build, 14.4 and 15.5 S, is identical in both arms.

Result 4: with more than one key the known-log base loses by a growing factor

Model built from the measured parts. The first key is the measured run. Each further key under the same G costs the relations phase per relation found: the abscissa matrix has been solved once, and the known-log base never needed one. S per key at `T` keys, ratio known-log over abscissa, 90% interval.

curve	w	per-key online limit, abscissa	per-key online limit, known-log	T = 1	T = 4	T = 16	T = 64
c1	0	0.040	4.13	0.97 [0.92, 1.03]	1.61 [1.42, 1.84]	4.08 [3.31, 4.95]	12.9 [10.0, 16.0]
c1	1	0.110	11.4	0.92 [0.82, 1.03]	2.11 [1.72, 2.56]	6.60 [5.11, 8.31]	21.2 [16.0, 27.0]
c2	0	0.036	3.94	0.97 [0.92, 1.02]	1.55 [1.35, 1.76]	3.80 [3.03, 4.62]	11.9 [9.15, 15.0]
c2	1	0.101	11.0	0.91 [0.81, 1.02]	2.02 [1.63, 2.45]	6.25 [4.74, 7.88]	20.3 [15.0, 25.8]

PLAN's E1 prediction that "as T grows both tend to the online limit and the ratio tends to 1" is also wrong. The online limits differ by about 100. In the same model the abscissa arm's amortised S per key is below the rho reference at T = 64 at every weight, and at T = 16 only at w = 1 (one G, 24 bits only); the known-log arm stays at 4 to 11 S per key and never gets there. Nothing here says anything about 192 or 224 bits.

What changes in PLAN

1. **§2 and E1 yield prediction.** Replace "ratio 1.00" by the sumset law: hit probability about `(2mB + 1) / r` for an arithmetic-progression base at `m = 2`, `(4B + 1) / r`. The interval [0.97, 1.03] cannot be met and should not be the pass condition.
2. **E1 "total S ratio < 1 by the set-up and LA share".** The measured share is 0% to 2.5%. The decision rule can still be run, but it will read "accounting" unless the arms differ in trials, which only a chance match of counts produced here.
3. **E1 multi-target block.** The ratio rises with T (about 12 to 21 at T = 64), it does not tend to 1.
4. **Family optimum.** `B ≈ r^(1/3)` is the optimum for an unstructured base. For the known-log base the pair table is mostly redundant, so the comparison that matches it is a baby-step table of `±[i]G` probed by one lookup per target (online cost about `r / 4B` targets), which is `bsgs.negation`. The next row should be known-log against that arm at equal memory, not against the abscissa base.
5. **N.** The known-log arm stops at its first relation, so each run adds one hit. A 90% interval of ±3% on its yield needs about 3000 hits, so about 3000 targets per curve, not 64. Either raise N, or let the known-log arm keep collecting relations after the first (a full-rank-style stop) so one run measures the rate.
6. **E2, E4, E3, E5.** Not run. E2′s N_0 argument is unchanged for the abscissa base; for the known-log base the same sumset collapse applies on top of it.

Checks

- All 288 records verified (`[d]G = Q` checked by the runner).
- `ic-knownlog` linear algebra: 0 group operations, 0 `row_ops`, rank 1, and GAE 0 in every record (`phases[linear_algebra]`).
- Base construction of the known-log arm: B − 1 charged additions (`known_log_additions`), 0.064 to 0.070 S.
- Unit tests, `src/cryptanalysis/ic_framework/mod.rs`: logs recovered on a 24-bit prime-order curve; digest identical for the same G and different for another; `±[i]G` closed under negation.
- Not done: E0 coupled-seed checks, `ic-abscissa-online`, A/A control, `m = 3`, the planted `B + 1` control, ecbench band against rho, any other bit size.
- Registry: the two curves are not in `docs/curves/registry.json` (AGENTS.md §11); register them before any citation in prose outside this scratch note.
- No canonical graph or leaderboard is changed: this is a scratch row and it lands no improvement (the known-log base is not faster than the baseline).

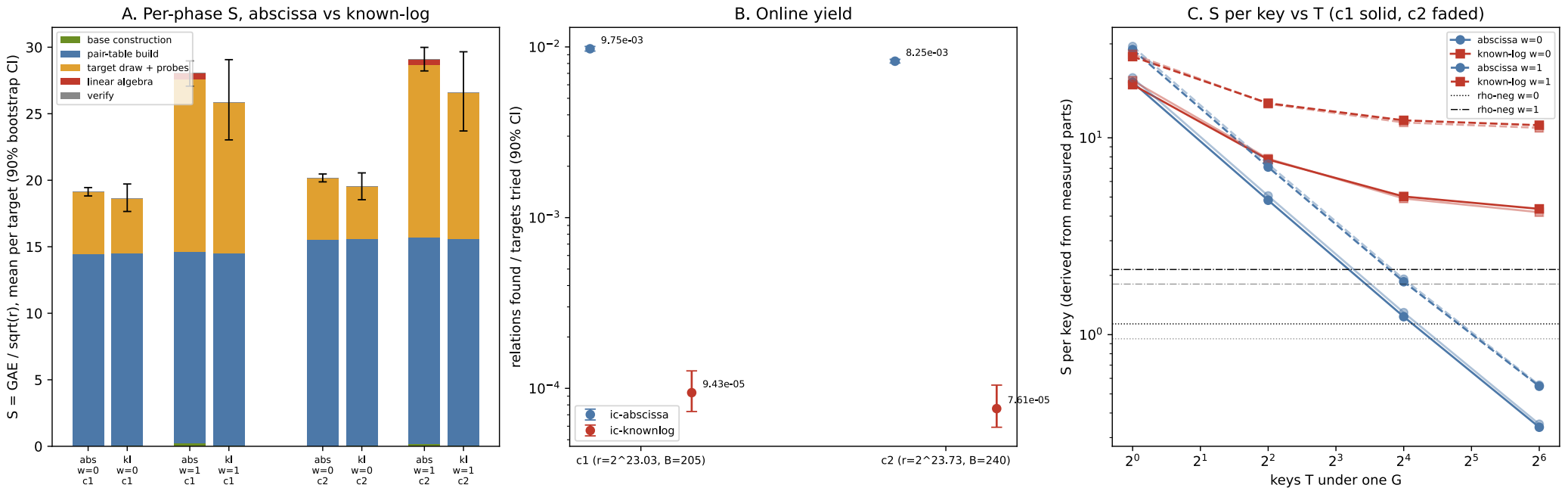


Figure 1: E1 dry run. A: mean S per target by phase (90% bootstrap interval on the total). B: relations found per target tried. C: S per key against keys T under one G, derived from the measured parts. Sessions ECBS1hdfb4b2694578 and ECBS1h3e4cb8782f3c, scratch dry run.